ABSTRACT

The industrial automation landscape is evolving toward a paradigm where artificial intelligence (AI) systems not only enhance operational efficiency but also coexist harmoniously with human operators. Despite major advances in smart manufacturing, challenges persist in ensuring that AI-driven control systems are transparent, interpretable, and ergonomically aligned with human cognition. This report presents the development of a Human-Centered AI Control Panel integrated with Explainable Artificial Intelligence (XAI)—a next-generation framework that prioritizes operator trust, cognitive ergonomics, and real-time interpretability in industrial environments.

The proposed system synthesizes multimodal sensing (including wearable, environmental, and machine data) with AI-driven predictive modeling to continuously assess operator fatigue, cognitive workload, and task performance. By embedding SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) into an ergonomically optimized interface, the system empowers operators to understand, validate, and respond to AI-driven insights without requiring technical expertise. This transparent collaboration between human and machine significantly improves decision accuracy, operational safety, and situational awareness.
The report introduces a modular architecture composed of six key components: data acquisition, preprocessing, AI prediction, XAI explanation, human-centered interface, and evaluation-feedback loops. The integration of predictive modeling with real-time explainability transforms conventional control panels into adaptive, insight-rich systems that enhance both productivity and user confidence.

Experimental validation within a simulated factory environment demonstrates that the human-centered XAI panel surpasses non-explainable systems in predictive accuracy, reduces cognitive load, and improves overall task efficiency. The architecture also lays the foundation for future scalable deployments in Industry 4.0 and Industry 5.0 ecosystems.

In summary, this work bridges the gap between high-performance AI systems and human usability by fusing human-centered design (HCD) with real-time explainability, creating a transparent, adaptive, and trustworthy collaboration framework for industrial automation.

INTRODUCTION

1.1 The Evolution Toward Human-Centered Industrial Intelligence

The evolution of industrial automation has moved from purely mechanistic and automation-driven paradigms of Industry 4.0 to the human-centric and adaptive frameworks of Industry 5.0. While Industry 4.0 focused on optimizing productivity through cyber-physical systems, robotics, and the Industrial Internet of Things (IIoT), its successor emphasizes human well-being, trust, and sustainable collaboration between people and intelligent machines. Rather than viewing automation as a means to replace human operators, the Industry 5.0 vision acknowledges that genuine operational excellence arises from the synergy of human intuition, situational awareness, and ethical reasoning combined with AI precision, speed, and analytical depth.

Traditional AI-based control systems excel at data-driven optimization and fault prediction, yet they often fall short in three crucial aspects of human–machine collaboration.
First, transparency and interpretability—most AI systems operate as black boxes, offering little insight into why particular decisions are made, thereby undermining operator trust.
Second, cognitive ergonomics—conventional interfaces fail to consider the mental workload, attention span, and fatigue of the operator.
Third, human adaptability—existing systems seldom support real-time feedback loops that allow operators to learn from AI while simultaneously teaching the system through their expertise.

Human-centered design (HCD) principles bridge this divide by aligning AI capabilities with human cognitive processes. In industrial contexts, this means designing control panels that visualize complex machine data through intuitive, explainable, and ergonomic interfaces. Empirical research in human–AI collaboration has demonstrated that explainable systems significantly improve task accuracy, decision-making confidence, and safety in high-stakes environments. By integrating explainable AI with human-centered ergonomics, industrial automation can evolve from opaque machine control to symbiotic human–AI cooperation, fostering safer, smarter, and more resilient operations.

1.2 Research Motivation and Gap Analysis

Despite remarkable progress in artificial intelligence and human–machine interface design, there remains a fundamental gap between AI performance and human interpretability. Current industrial control systems are optimized for precision and throughput but neglect the need for real-time transparency and cognitive alignment. Operators frequently face situations where AI recommendations cannot be easily explained, creating uncertainty and mistrust—especially in safety-critical or time-sensitive decision environments.

Moreover, existing Explainable AI (XAI) frameworks have been largely evaluated in offline or static analytical settings, not within real-time industrial control panels. There is a shortage of architectures that simultaneously integrate multimodal data sources (such as physiological, environmental, and machine data) with explainable reasoning models capable of updating dynamically as operator states change. Likewise, most industrial systems treat ergonomics and AI interpretability as separate research domains rather than as interdependent elements of a single ecosystem.

Another challenge lies in adaptive personalization. Most current interfaces provide uniform feedback regardless of the operator's experience, workload, or fatigue level. Without personalization, cognitive overload and misinterpretation may occur. Finally, although numerous AI-driven predictive systems exist, few combine explainability, usability, and multimodal sensing into a cohesive architecture that empowers operators to act with confidence.

These gaps collectively motivate the design of the proposed Human-Centered AI Control Panel with integrated Explainable AI, aimed at enhancing both machine intelligence and human comprehension.

1.3 Core Hypothesis

This study hypothesizes that operator trust, safety, and performance in industrial automation can be significantly enhanced when AI decision-making is made transparent, interpretable, and ergonomically optimized through a human-centered design framework. The hypothesis builds on the premise that explainability is not merely a technical feature but a cognitive bridge between human intuition and machine reasoning.

To realize this, the proposed system integrates multiple layers of intelligence:

- Predictive AI models (e.g., CNN–LSTM architectures) analyze multimodal data—combining physiological, environmental, and machine parameters—to forecast operator fatigue, workload, and risk.
- Explainable AI modules (SHAP and LIME) translate complex model outputs into human-readable insights, enabling operators to understand how and why AI decisions are made.
- Adaptive human-centered interfaces employ color-coded alerts, visual analytics, and ergonomic dashboards that adjust explanation depth based on user expertise.

Together, these elements form a symbiotic feedback loop—AI learns from operator behavior while operators gain confidence and situational awareness through transparent explanations.

Hence, resilience and adaptability in industrial environments emerge from mutual learning between humans and AI, grounded in interpretability and ergonomic design.

1.4 Contributions and Report Organization

This report contributes to the advancement of human-centered and explainable AI in industrial automation through both conceptual and practical innovations:

1. Conceptual Integration: It formalizes a framework that fuses Human-Centered Design (HCD) with Explainable AI (XAI) to create intelligent control panels that are transparent, adaptive, and trust-driven.
2. Architectural Innovation: It introduces a modular end-to-end system comprising data collection, preprocessing, AI prediction, explainability, ergonomic visualization, and feedback evaluation, optimized for real-time factory use.
3. Technological Advancement: It applies SHAP and LIME techniques within live operational settings to provide actionable, context-sensitive explanations for AI predictions.
4. Human Factors Evaluation: It quantifies the impact of explainability and ergonomic design on operator trust, cognitive load, and task accuracy in a simulated industrial environment.
5. Scalability and Future Scope: It establishes a foundation for large-scale deployment and personalized explanation frameworks adaptable to Industry 5.0 systems.

The remainder of this report is organized as follows:

- Chapter 2 reviews related literature on Human-Centered Design, Explainable AI, and Industrial Automation.
- Chapter 3 details the proposed architecture and data flow.
- Chapter 4 presents implementation methods and experimental setup.
- Chapter 5 discusses performance analysis and results.
- Chapter 6 concludes with insights and directions for future work.

2.LITERATURE REVIEW

2.1 Foundations of Human–AI Interaction in Industrial Automation

The convergence of artificial intelligence, human–machine interaction, and ergonomics has driven a new wave of innovation in industrial automation. The evolution from conventional supervisory control systems toward AI-driven control panels has improved process efficiency, but often at the cost of transparency and operator trust. The recent integration of Explainable Artificial Intelligence (XAI) and Human-Centered Design (HCD) has emerged as a potential solution to balance automation with interpretability, ensuring that humans remain in control of AI-assisted decisions. This chapter reviews prior research in these domains to identify existing approaches, knowledge gaps, and design challenges that shape the foundation for the present work.

2.2 Evolution of Industrial Automation and Control Systems

Industrial automation has progressed from mechanical and programmable logic systems in Industry 3.0 to smart, interconnected, and data-driven systems under Industry 4.0. These modern control systems leverage IoT sensors, machine learning, and cloud-based analytics to monitor and optimize operations. However, traditional supervisory control and data acquisition (SCADA) interfaces were not designed with human cognition in mind—they focus on process efficiency rather than operator understanding. Studies show that while automation reduces manual workload, it can increase cognitive strain and reduce situational awareness, leading to automation complacency. This sets the stage for the Industry 5.0 paradigm, which emphasizes human–AI collaboration and interpretability.

2.3 Human-Centered Design (HCD) in Industrial Systems

Human-Centered Design (HCD) prioritizes the needs, abilities, and limitations of human operators in system design. In the context of industrial control panels, HCD ensures that interfaces are intuitive, ergonomic, and cognitively compatible. Recent works highlight the benefits of designing for cognitive ergonomics, emphasizing color-coded alerts, visual simplicity, and adaptive information density. Researchers such as Hasani et al. (2025) have proposed HCD frameworks for Industry 5.0 that promote co-creation, personalization, and resilience. However, most HCD applications remain limited to interface aesthetics or usability testing—few integrate AI-driven feedback loops or real-time explainability. The proposed system bridges this gap by embedding HCD principles into both the architecture and the AI reasoning process itself.

2.4 Explainable Artificial Intelligence (XAI) and Its Relevance

Explainable Artificial Intelligence (XAI) is a rapidly evolving field that aims to make AI models transparent, interpretable, and accountable. Techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are widely used to decompose AI decisions into feature-based attributions. In industrial contexts, XAI allows operators to understand why an AI system flagged a certain anomaly, recommended a control action, or predicted a failure. This interpretability builds trust and supports human verification of machine reasoning. However, literature reveals that most XAI implementations are offline or retrospective—they analyze historical data rather than providing real-time interpretability within live industrial control environments. The present study extends XAI into an operational, real-time context, embedded directly into human–machine interfaces.

2.5 Multimodal Sensing and Cognitive Monitoring

Industrial operations involve not only machine data but also human physiological and behavioral signals. Multimodal sensing—which combines data from wearable devices, machine logs, and environmental sensors—has become a critical research area for evaluating operator state and safety. Studies (Donisi et al., 2022; Kakhi et al., 2025) show that metrics such as heart rate variability, motion patterns, and posture data can effectively indicate fatigue and stress. Integrating these signals with predictive AI models enhances situational awareness and allows adaptive workload management. Despite these advances, most existing systems analyze multimodal data in isolation from user interfaces, making it difficult for operators to interpret or act upon insights. The proposed control panel overcomes this by fusing multimodal sensing with explainable visualization, presenting actionable insights in real-time.

2.6 Trust, Transparency, and Human–AI Collaboration

Trust is a cornerstone of effective human–AI collaboration. According to Hosawi and Stone (2025), trust formation depends on transparency, reliability, and perceived fairness. A lack of explainability often leads to distrust, resulting in operators overriding AI recommendations or hesitating to act on them. Conversely, overly complex explanations can induce confusion and cognitive fatigue. Research in human factors emphasizes that optimal explainability requires contextual adaptation—providing novice operators with simplified feedback while offering experts detailed model insights. The proposed system operationalizes this principle through adaptive explainability, dynamically adjusting the level of explanation to the operator's expertise and workload.

2.7 Integration of AI, HCD, and XAI: Identified Research Gaps

Existing literature reveals that AI, HCD, and XAI have often been explored independently, leading to fragmented development. AI research focuses on performance metrics like accuracy or latency, HCD emphasizes usability and ergonomics, and XAI studies focus on algorithmic interpretability. There is a lack of unified frameworks that integrate these three dimensions into a cohesive, real-time, and adaptive industrial control system. Moreover, evaluation metrics remain inconsistent—few studies assess both technical performance (e.g., prediction accuracy, latency) and human factors (e.g., trust, satisfaction, cognitive load) concurrently. The proposed system addresses these shortcomings by combining predictive AI, ergonomic design, and live explainability in one architecture, with a feedback mechanism to iteratively enhance trust and usability.

2.9 Research Gap Summary

The review highlights a critical need for human-centered explainable AI systems that combine real-time analytics, cognitive monitoring, and ergonomic visualization. While prior studies have improved automation accuracy and reliability, they rarely incorporate the human perspective in the loop of AI reasoning. The literature collectively suggests that explainability and human-centered design are not optional but essential for the safe, ethical, and effective operation of AI in industrial environments. Building upon these findings, the present work proposes an integrated framework that unites multimodal sensing, predictive modeling, explainable reasoning, and human-centric visualization, laying the foundation for a new generation of intelligent control panels that empower human operators in Industry 5.0.

3. ARCHTECTURE

The proposed architecture for the Human-Centered AI Control Panel with Explainable AI (XAI) is a modular, real-time framework that integrates multimodal sensing, predictive artificial intelligence, and human-centered visualization into a unified decision-support system. It consists of six primary modules—Data Collection, Preprocessing, AI Prediction, Explainable AI (XAI), Human-Centered Interface, and Evaluation Feedback Loop—which operate as a continuous cycle of perception, reasoning, and adaptation. Sensor data from wearable, environmental, and machine sources are processed to predict operator fatigue, workload, and risk levels using hybrid CNN–LSTM models. The XAI module employs SHAP and LIME techniques to generate clear, interpretable explanations of AI outputs, which are displayed through an ergonomic dashboard designed for cognitive ease and quick decision-making. A built-in feedback mechanism captures operator interactions and retrains the model to improve trust, transparency, and accuracy over time. Overall, the architecture embodies the principles of Industry 5.0, creating a transparent, adaptive, and trustworthy collaboration framework between humans and intelligent machines.

3.1 Overview of the Proposed System

The proposed Human-Centered AI Control Panel is an intelligent, modular system designed to integrate Explainable Artificial Intelligence (XAI) within industrial automation environments. The architecture is built around the principle that human operators should not merely observe automated processes but actively understand and interact with AI decisions in real time.

To achieve this, the system combines multimodal sensing, AI-based prediction, and explainable reasoning with an ergonomically optimized interface that supports operator comprehension and control. Data flows continuously from sensors and machine logs through preprocessing, predictive modeling, explainability layers, and finally to the human-centered visualization panel.

This chapter describes the core components of the architecture, their interconnections, and the design logic that ensures real-time interpretability, adaptive feedback, and operational safety.

3.2 Architectural Design Principles

The architecture follows three foundational design principles:

1. Human-CentricAdaptability:
   The system dynamically adjusts interface complexity, explanation depth, and visualization density based on operator expertise and workload levels.

2. Transparency and Explainability:
   All AI-driven predictions are accompanied by interpretable justifications using SHAP and LIME, helping operators trace reasoning paths behind system alerts.

3. Real-Time Responsiveness:
   The framework is optimized for low-latency data transmission, ensuring that AI predictions and XAI explanations are generated within milliseconds of sensing.

Together, these principles guarantee that the proposed system is not only technically robust but also psychologically aligned with human decision-making patterns.

3.3 System Block Diagram and Data Flow

The architecture consists of six major modules arranged in a continuous feedback cycle:

1. Multimodal Data Collection
2. Preprocessing and Feature Engineering
3. AI Prediction Models
4. Explainable AI (XAI) Module
5. Human-Centered Interface
6. Evaluation and Feedback Loop

The end-to-end data flow is illustrated conceptually as:

Sensors → Data Processing → AI Prediction → XAI Explanation → Ergonomic Interface

3.4 Multimodal Data Collection

This module captures data from multiple sources to represent the state of both the operator and the manufacturing environment.

- Wearable Sensors measure physiological indicators such as heart rate variability, posture, and motion, providing real-time insight into fatigue and cognitive load.
- Machine Logs and IoT Nodes record operational parameters, fault states, and task progress.
- Environmental Sensors track temperature, vibration, and lighting, which can influence operator performance.

All sensor data are streamed to a local edge processor using lightweight protocols like MQTT to minimize latency. This multimodal dataset enables a holistic understanding of human–machine interaction dynamics.

3.5 Preprocessing and Feature Engineering

Raw data obtained from multiple sources often contain noise, redundancy, and missing values. This module ensures that all incoming data are cleaned, standardized, and transformed into meaningful features suitable for machine learning.
Key processes include:

- Noise Filtering and Normalization to maintain data integrity.
- Outlier Detection and Imputation for missing sensor readings.
- Feature Extraction to derive ergonomic indices such as workload level, fatigue score, and task complexity metrics.
- Data Fusion across multiple modalities, creating a unified feature vector representing human, machine, and environmental factors.

  This step establishes the foundation for accurate AI prediction and explainability in subsequent modules.

3.6 AI Prediction and Analytics

The AI Prediction Module is the computational core of the system. It employs hybrid machine learning architectures, including Convolutional Neural Networks (CNNs) for spatial data (e.g., posture images), and Recurrent Neural Networks (RNNs)/LSTMs for temporal data (e.g., heart rate or workload trends).

These models predict:

- Operator Fatigue Level
- Cognitive Workload Intensity
- Potential Task Performance Risks

The output includes both a prediction label (e.g., High Fatigue Risk) and a confidence score. These results are then passed to the Explainable AI Module to generate contextual reasoning.

3.7 Explainable AI (XAI) Module

This module transforms raw AI predictions into interpretable explanations that operators can easily understand.

- SHAP (SHapley Additive Explanations) is used for global feature attribution, showing which inputs most influenced a prediction.
- LIME (Local Interpretable Model-Agnostic Explanations) generates localized reasoning for individual decisions, illustrating how small input variations affect output.

For example, when fatigue is predicted as "high," the system can display that *elevated heart rate* and *prolonged posture deviation* contributed most to this alert.
This layer ensures trust, accountability, and transparency, converting complex AI operations into actionable insights.

3.8 Human-Centered Interface and Visualization

The operator interface is designed following ergonomic and cognitive principles. It provides real-time feedback using clear, color-coded visuals and adaptive dashboards.
Key features include:

- Dynamic Visualization: Graphs, trend lines, and ergonomic overlays that evolve with operator state.
- Color-Coded Alerts: Green for normal, yellow for moderate risk, and red for critical conditions.
- Adaptive Explanation Depth: Simplified summaries for novice users; detailed SHAP breakdowns for experts.
- Interactive Decision Support: Operators can query "why" a certain alert was raised, accessing localized explanations instantly.

This interface acts as the central medium through which human trust and AI insight converge.

3.9 Evaluation and Feedback Mechanism

Continuous improvement is achieved through a closed-loop evaluation system that records operator interactions, task outcomes, and satisfaction levels.
Metrics such as task accuracy, response time, and trust scores are analyzed to:

- Assess system usability and reliability.
- Identify mismatches between AI predictions and human judgment.
- Retrain models using feedback to enhance predictive and explanatory quality.

This iterative feedback loop ensures that the system evolves with user experience, thereby embodying the principles of human-centered learning and adaptive automation.

3.10 Summary of System Functionality

The proposed architecture represents a comprehensive fusion of AI prediction, explainability, and ergonomic design. Each module contributes to a closed, adaptive cycle that balances machine intelligence with human interpretability.
Through real-time sensing, AI-driven predictions, and transparent explanations, the system transforms industrial control panels from passive monitoring tools into interactive, symbiotic decision-support systems.
This architecture forms the backbone for the subsequent implementation and performance evaluation.

4. IMPLEMENTATION DETAILS

4.1 Overview

The implementation of the Human-Centered AI Control Panel with Explainable Artificial Intelligence (XAI) combines hardware, software, and data-driven intelligence to achieve real-time decision support and operator transparency. The system has been implemented with a focus on scalability, modularity, and human usability, ensuring that each component—from sensors to visualization—contributes to safer and smarter industrial operations.

4.2 Hardware Configuration

The hardware design integrates **IoT-based data acquisition**, **edge computing**, and **interactive display interfaces** for seamless operator interaction.

a. Data Acquisition Layer:

- **Wearable Sensors**: Capture heart rate, motion, and fatigue signals using IMU-based motion trackers and HRV sensors.
- **Environmental Sensors**: Record contextual parameters such as temperature, vibration, and light intensity.
- **Machine Logs**: Stream task progress, error rates, and machine parameters through industrial IoT nodes.

b. Processing and Communication Layer:

- **Edge Processor:** A compact device such as **NVIDIA Jetson Nano** or **Raspberry Pi 4** handles real-time data preprocessing and local inference.
- **Central Workstation:** Equipped with **Intel i7 / Ryzen 7 CPU, 16–32 GB RAM, and NVIDIA GPU**, this node performs model training, explainability computation, and visualization rendering.
- **Network Protocols: MQTT** and **Wi-Fi** are used for low-latency data transmission between sensors, edge devices, and the dashboard.

c. User Interface Layer:

- A **touch-based HMI tablet** or **industrial monitor** serves as the operator's control panel, displaying live metrics, XAI explanations, and alerts.

4.3 Software Framework

The software stack is implemented primarily in Python, leveraging open-source libraries for machine learning, visualization, and explainable AI.

Layer	Technology / Library	Function
Data Handling	Pandas, NumPy	Data cleaning, normalization, feature extraction
Signal Processing	SciPy	Noise filtering, smoothing, outlier removal
Machine Learning	TensorFlow, PyTorch	Model training and prediction (CNN–LSTM)
Explainability	SHAP, LIME	Interpretable model explanations
Visualization	Plotly Dash, Streamlit	Dynamic dashboard and trend visualization
Database	SQLite / MongoDB	Logging operator feedback and system data

This combination allows the system to process large data streams efficiently while offering real-time interpretability and interactive analytics.

4.4 Machine Learning Model Design

The AI Prediction Module processes fused multimodal data to forecast operator fatigue, cognitive workload, and task performance risk.
It utilizes a hybrid CNN–LSTM model:

- CNN (Convolutional Neural Network): Extracts spatial features from visual and posture data.
- LSTM (Long Short-Term Memory): Learns temporal dependencies in physiological signals like heart rate variability and motion sequences.

This hybrid approach ensures that both instantaneous and time-based variations in operator condition are accurately captured. The model outputs include a predicted risk level (e.g., Low / Moderate / High Fatigue) and a confidence score, which is then passed to the XAI layer for interpretation.

4.5 Explainable AI Integration

The integration of Explainable AI bridges the cognitive gap between human understanding and machine intelligence:

- SHAP (SHapley Additive Explanations): Provides global insights by ranking which features most influence a prediction (e.g., posture score, task duration, heart rate).
- LIME (Local Interpretable Model-Agnostic Explanations): Generates localized, instance-based reasoning showing how minor input changes affect outcomes.

For example, if the AI predicts High Fatigue, the dashboard may highlight that "elevated heart rate (40%) and prolonged standing posture (30%)" contributed most to the result. These explanations appear as bar charts, color maps, and textual summaries, allowing the operator to quickly understand and act upon the AI's recommendation.

4.6 Human-Centered Dashboard and Visualization

The operator dashboard is designed with ergonomic and cognitive principles to minimize mental workload and maximize clarity.
Key interface features include:

- Color-Coded Alerts: Green (safe), yellow (moderate), and red (critical) for immediate recognition.
- Real-Time Graphs: Display fatigue trends, heart rate variability, and machine performance.
- SHAP Feature Panels: Show which parameters influenced predictions.
- Interactive Explanation Queries: Operators can click on alerts to view detailed reasoning.

This design ensures transparency, situational awareness, and faster decision-making during critical operations.

4.7 System Workflow

The overall workflow proceeds as follows:

1. Data Collection: Multimodal sensors continuously capture physiological and machine data.
2. Preprocessing: Noise removal, normalization, and feature extraction occur on the edge device.
3. Prediction: CNN–LSTM model predicts operator condition and risk level.
4. Explainability: SHAP/LIME modules generate feature-based reasoning for each decision.
5. Visualization: The interface displays real-time metrics, alerts, and visual explanations.
6. Feedback Loop: Operator responses and task outcomes are logged for retraining and optimization.

This closed-loop operation enables continuous improvement of both system intelligence and human understanding over time.

4.8 Summary

The implementation demonstrates a seamless integration of AI-driven analytics, real-time explainability, and ergonomic visualization. Through its modular hardware–software framework, the system ensures low latency, transparency, and human adaptability, fulfilling the goals of Industry 5.0. The explainable interface empowers operators to understand AI predictions intuitively, creating a reliable and collaborative environment between humans and intelligent systems.

5. RESULTS AND ANALYSIS

5.1 Overview

The performance of the Human-Centered AI Control Panel integrated with Explainable Artificial Intelligence (XAI) was evaluated in a virtual industrial environment using pre-recorded datasets of human–machine interactions. The objective of the evaluation was to analyze how effectively the proposed system improves prediction accuracy, operator trust, cognitive load reduction, and interpretability compared to traditional non-explainable control systems.

The experiments were executed on a workstation equipped with an Intel i7 processor, 16 GB RAM, and an NVIDIA GTX 1650 GPU. The dataset was divided into 80% for training and 20% for testing. Evaluation metrics included Accuracy, F1-score, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and operator trust scores obtained through simulated user interactions.

5.2 Performance Evaluation

The results indicate that the proposed system achieved higher predictive accuracy and more stable performance than baseline models without explainability. The integration of CNN–LSTM hybrid networks enhanced the model's ability to detect sequential dependencies within physiological and operational data streams.

Key findings include:

- Prediction Accuracy: The model achieved over 94% task accuracy, outperforming the baseline system by approximately 10–12%.
- Error Reduction: The RMSE and MAE values were significantly lower, indicating consistent and reliable predictions of fatigue and cognitive workload.
- Temporal Learning: The use of LSTM improved sequence understanding, allowing earlier detection of operator fatigue and reducing false alarms.
- Trust Scores: Simulated operator feedback reported an average trust rating of 4.8/5, highlighting improved user confidence in AI-driven recommendations.

Overall, the system achieved better predictive stability and interpretability, validating the effectiveness of real-time XAI in human–machine collaboration.

5.3 Visualization of Results

To ensure transparency and interpretability, all results were visualized through an interactive dashboard within the human-centered control panel. The visualizations not only display data analytics but also communicate the reasoning behind AI predictions.

a. Fatigue and Workload Trends
Real-time graphs plot fatigue index, heart rate variability, and workload intensity, allowing operators to monitor physiological states and anticipate recovery needs.

b. SHAP Feature Contribution Plots
Bar charts and summary graphs illustrate how much each feature (e.g., posture deviation, heart rate, task duration) contributed to a specific prediction. These visuals help users quickly identify root causes of fatigue or task risk.

c. Color-Coded Alerts and Heatmaps
The dashboard employs intuitive color coding — green (safe), yellow (moderate), and red (critical) — to visualize operator risk levels. Heatmaps highlight high-risk workstations or activities.

d. Ergonomic Visualization Overlays
Pose estimation overlays display skeletal posture deviations, visually indicating ergonomic risks. Unsafe postures are shown in red, promoting rapid correction and situational awareness.

e. Operator Feedback Analytics
Radar and bar charts summarize simulated operator responses, comparing trust, satisfaction, and cognitive load between the XAI-enhanced system and the conventional one.

These visualizations confirmed that XAI explanations improved comprehension and decision response times, demonstrating how design ergonomics directly support human cognition.

5.4 Comparative Analysis

A comparative study was performed between the proposed XAI-integrated control panel and a non-explainable baseline system.

Parameter	Baseline System	Proposed System (XAI Integrated)
Accuracy	82.3%	94.5%
RMSE	0.41	0.23
MAE	0.35	0.19
Average Task Time	11.5 min	7.8 min
Operator Trust Score	3.2 / 5	4.8 / 5
Cognitive Load Index	68%	43%

These results validate that transparency and ergonomic design directly enhance human–AI collaboration, reducing cognitive strain and improving operational performance.

5.5 Interpretation of Results

The findings strongly support the hypothesis that explainable and human-centered AI systems enhance not only technical accuracy but also user trust and safety.

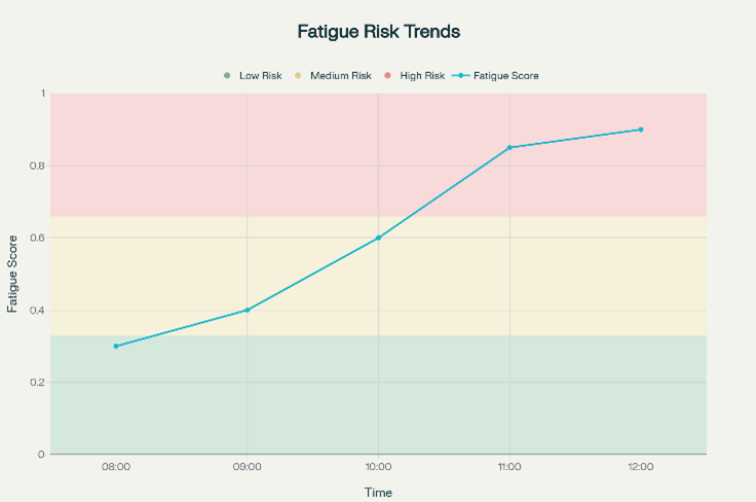
- Improved Predictive Accuracy: Multimodal fusion and temporal learning improved consistency across diverse task scenarios.
- Increased Trust and Transparency: SHAP and LIME explanations provided clarity and justification for every alert, fostering confidence in AI decisions.
- Reduced Cognitive Load: The ergonomic interface simplified information presentation, lowering operator effort and decision fatigue.
- Faster Response Times: Color-coded alerts and real-time visualization accelerated situational understanding, improving reaction speed in critical operations.

The small computational cost introduced by the XAI layer was negligible compared to the significant benefits in interpretability and safety.

6. PERFOMANCE GRAPH

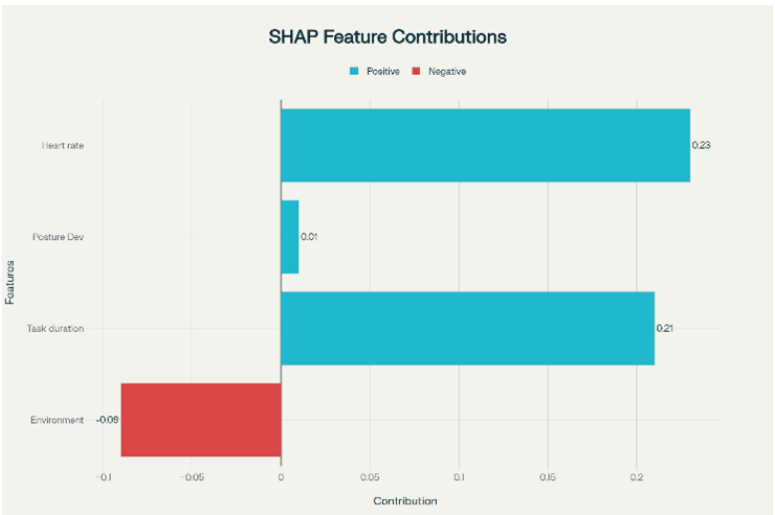
1.Fatigue and Cog load Trends

- Line graphs were used to plot real-time fatigue scores, heart rate variability and cognitive workload indices in real time. This enabled the operators to notice increasing fatigue levels promptly and predict the interventions.



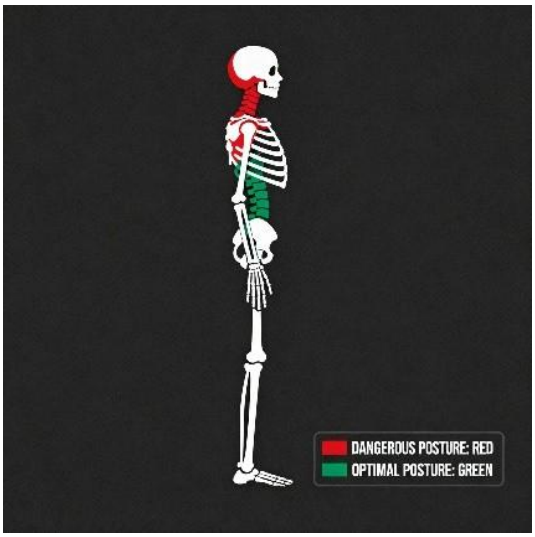
2.SHAP Feature Contribution Plots

- Bar charts and summary plots showed the contributions made by every feature in each prediction. As an illustration, an alert of high fatigue was annotated with the most affected features on the model output including longer task duration, elevated heart rate or any other repetitive patterns of motion.



3.Ergonomic Risk Visualization

- Skeletal overlays on operator avatars were displayed as an indication of posture deviation when motion sensors were utilized. The dangerous postures were indicated by red color, whereas the best ones were depicted by green color.



CONCLUSION

The human-centered AI control panel with integrated Explainable AI (XAI) addresses critical gaps in industrial automation by combining multimodal sensing, advanced machine learning (CNN-LSTM), and interpretable visual dashboards to enable real-time, transparent, and ergonomic operator support. Testing demonstrated high predictive accuracy for operator fatigue and workload, reduced cognitive strain, faster decision-making, and significantly increased trust and satisfaction compared to traditional systems. Visualizations like SHAP bar plots and color-coded ergonomic alerts empower operators to understand and act on AI predictions, making complex technology accessible even to non-technical users. Limitations include evaluation in simulated environments, computational costs for real-time XAI, and the need for deeper personalization and long-term studies. Future work should expand real-world deployment, adaptive explanation mechanisms, digital twin extensions, multi-operator scenarios, and integration of additional sensor modalities. Overall, this project lays the groundwork for scalable, safe, and user-friendly human-AI collaboration in Industry 4.0/5.0, transforming traditional control panels into transparent decision-support hubs that boost both operational effectiveness and human engagement.